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15 February 1952
SUPERSEDING
MIL-C-10146(SigC)
17 February 1950

MILITARY SPECIFICATION

CASE, CRYSTAL HOLDER (CS-137)

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments.

1. SCOPE

1.1 This specification covers one type of case for crystal units, intended for use in carrying 120 crystal holders FT-243.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications, standards, and drawings, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

MILITARY

JAN-P-105 —Packaging and Packing for Overseas Shipment—Boxes, Wood, Cleated, Plywood.

JAN-P-106 —Packaging and Packing for Overseas Shipment—Boxes; Wood, Nailed.

JAN-P-120 —Packaging and Packing For Overseas Shipment — Cartons, Folding, Paperboard.

JAN-P-125 —Packaging and Packing for Overseas Shipment—Barrier-Materials, Waterproof, Flexible.

JAN-P-133 —Packaging and Packing for Overseas Shipment—Boxes, Set-Up, Paperboard.

JAN-P-140 —Packaging and Packing for Overseas Shipment — Adhesives, Water-Resistant, Case-Liner.

U. S. ARMY

72-53 —Finishes (For Ground Signal Equipment).

STANDARDS

MILITARY

MIL-STD-105—Sampling Procedures and Tables for Inspection By Attributes.

MIL-STD-129—Marking of Shipments.

DRAWINGS

SIGNAL CORPS

SC-D-964 —Slide Catch Assembly and Details.

SC-D-11553 —Case CS-137, Assembly.

SC-D-11554 —Case CS-137, Details.

SC-D-11555 —Case CS-137, Details.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Construction.—The equipment shall be constructed in accordance with Drawings SC-D-964, SC-D-11553, SC-D-11554, and SC-D-11555, and shall meet the inspection and tests specified in section 4.

3.2 Preproduction samples.—The contractor shall assemble two samples of the case

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CS–137 using the tools and methods similar to those which will be used in quantity production. These samples, together with test data showing complete compliance with this specification, shall be submitted to the designated Government agency, as specified (see 6.1). Approval of the submitted samples shall not be construed as authorizing any deviation from this specification.

3.2.1 *Approval.* — Preproduction samples shall be inspected and tested by the Government inspector for conformance with all specification requirements. If any of the samples are disapproved, the contractor may be required to submit additional samples until satisfactory samples have been submitted. Such additional samples shall be accompanied by a description of the changes which have been incorporated in the new samples in order to correct the faults of the rejected samples.

3.3 **Parts, materials, and processes.**—All parts, materials, and processes, not specifically specified herein shall be in accordance with the applicable specifications and drawings listed in section 2.

3.4 **Metals.**—The metals used shall be of the proper alloy and hardness necessary to provide the required strength and rigidity. Materials shall be of corrosion-resisting types or shall be suitably processed to resist corrosion. Dissimilar metals used in contact with each other shall be carefully selected in accordance with Specification 72–53 to prevent electrolytic corrosion in the presence of moisture.

3.5 **Welding.**—Welding shall be done in such a manner that thorough penetration and good fusion are secured, with welds free from scabs, blisters, pockmarks, bruises, or other injurious surface defects. From time to time, as the inspector may require, test pieces shall be prepared (representative of production) and welded by the process used in production. These test samples shall be examined for evidence of burning or other harmful effect due to the welding process and then stressed to failure in an approved testing machine. If failure occurs outside the weld, it may be assumed that the welding process, as represented in the prepared test pieces, is satisfactory. If failure occurs within the weld, or there is other evidence of weakness, the contractor shall make the necessary corrections in production.

3.6 **Tack and spot welds.**—Tack and spot welds may be used where dependency for strength of the weld is not placed on a single tack or spot weld.

3.7 **Cleaning.** — The equipment shall be thoroughly cleaned and shall be free of foreign material. After metal parts have been fabricated, they shall be cleaned in accordance with the best commercial practice. Welds shall be thoroughly cleaned of flux and oxidation products. All traces of corrosive compounds shall be removed to prevent subsequent corrosion of the parts, or of other parts adjacent thereto.

3.8 **Interchangeability.**—All parts of each case shall be mechanically interchangeable with all corresponding parts of the equipments on order.

3.9 **Ruggedness and reliability.**—All parts and details of construction of this case shall have the maximum of ruggedness and reliability. The case shall withstand long periods of repeated rough handling in transportation and use with a minimum of damage to itself and its contents.

3.10 **Workmanship.** — All parts shall be manufactured in a thoroughly workmanlike manner.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Contractor's inspection.** — Materials, parts, and complete equipments, before being offered for acceptance on contract, shall be inspected by the contractor for compliance with the applicable specifications. Materials

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and parts, before being assembled into the equipment, shall be inspected to the extent necessary to insure compliance with the applicable specifications. Items covered by subsidiary specifications shall be inspected and tested in accordance therewith. Inspection of materials, parts, and complete equipments shall take place at the point of manufacture or of assembly, as specified (see 6.1). If a Government inspector is not assigned to observe such inspection, the contractor may be required to furnish satisfactory evidence of proper inspection, such as certificates of compliance or certified test data.

4.1.1 *Government inspectors.*—Authorized Government inspectors shall be privileged to observe the fabrication, processing, and assembly of all materials, parts, and completed equipment used on the contract and to witness and participate in the contractor's inspection of such items.

4.2 **Government inspection.**—In order to facilitate Government inspection, the contractor shall submit the equipment in discrete lots if so required by the contracting officer. The size of each lot shall be as designated by the contracting officer, and acceptance of the lot shall be contingent upon results of Government inspection of samples from the lot.

4.3 **Production Inspection.**—Production inspection shall consist of visual and mechanical inspection (see 4.3.1). At the discretion of the contracting officer, production inspection may be conducted by either or both of the following methods:

100 percent inspection and nondestructive testing.

Statistical sampling with an acceptable quality level (AQL) as specified, (see 6.1). Standard MIL–STD–105 shall be used for statistical sampling, when applicable.

4.3.1 *Visual and mechanical inspection.*—Each case shall be inspected to ascertain that the materials and construction are in accordance with the drawings listed in section 2. Dimensional measurements shall be made on random samples or on each case at the discretion of the inspector. All catches, hinges, and covers shall be inspected for any scraping or binding.

4.3.2 *Test for proper application of paint.*—At the discretion of the inspector, painted surfaces of several sample chests selected from production shall be checked for satisfactory application of paint by testing the adhesion of the paint film in accordance with Specification 72–53.

5. PREPARATION FOR DELIVERY

5.1 **Preservation and packaging.**

5.1.1 *For domestic shipment.*—Commercial preservation and packaging will be acceptable.

5.1.2 *For overseas shipment.*

5.1.2.1 *Case CS–137.*—Package each case for mechanical and physical protection only, as follows: Wrap each case in flexible corrugated paper, securing the wrapping with gummed kraft tape. Place each wrapped case within a close-fitting folding carton or set-up box, conforming to Specifications JAN–P–120 and JAN–P–133, respectively. Seal the entire closure with gummed kraft tape.

5.2 **Packing.**

5.2.1 *For domestic shipment.*—Cases CS–137 shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, in such a manner as to insure acceptance and safe delivery to the designated point by common or other carriers, at the lowest rate.

5.2.2 *For overseas shipment.* — Place a quantity of case CS–137, packaged as specified in 5.1.2.1, within a wood cleated plywood or nailed wood box, conforming to specifications JAN–P–105 and JAN–P–106, re-

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spectively. Fabricate the boxes to fit the contents snugly. Each shipping container shall be lined with a waterproof bag, fabricated of waterproof barrier material, conforming to Specification JAN–P–125, for case liners. Closure shall be completely sealed with a continuous seam at least ¾ inch wide of water resistant adhesive complying with Specification JAN–P–140. Close and strap the shipping container as specified in the appendix of the applicable box specification. The maximum gross weight of any shipping container shall not exceed approximately 150 pounds.

5.3 Marking for shipment.

5.3.1 *Interior packages.* — Packages shall be marked in accordance with Standard MIL–STD–129.

5.3.2 *Shipping containers.*—In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Standard MIL–STD–129.

6. NOTES

6.1 Ordering data. — Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Agency to which preproduction samples are to be submitted (see 3.2).

(c) Place of final inspection (see 4.1).

(d) Acceptable quality level (see 4.3).

(e) Whether domestic or overseas shipment is required (see 5.2).

Notice—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

Custodian:
Army—Signal Corps

Other interest:
Army—T
Air Force.

McGregor & Werner, Inc.
Dayton, O. 600, 13 June 52

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